FWB intervals for conditional infit MSQ

Coverage of the population MSQ, item by item and simultaneously

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Purpose

An effect-size floor on infit judges each item by an interval for its population MSQ against a band of practical fit, $[1-\delta, 1+\delta]$, instead of testing MSQ = 1. That needs an interval for the observed MSQ under the data as they are, not under the null. The fractional weighted bootstrap (FWB) gives every respondent a random positive weight with mean 1, refits the model by CML with those weights, and recomputes infit. Its percentile interval reflects the sampling variability of the observed MSQ, including the re-estimation of the thresholds, and nobody drops out of a draw, so no category empties.

This note checks whether the intervals cover the population MSQ for three kinds of item: items under a true Rasch model, misfitting items, and fitting items distorted by the cascade when other items misfit. It also checks simultaneous intervals that use one max-type constant over all items, the infit analogue of Q3max*. The simulation is `fwb_msq_coverage.R`.

Design

- **Datasets.** Regenerated from `conditional_rmsd_partA.qmd` with its seeds: 20 dichotomous items evenly spaced on [−2, 2], theta ~ N(0, 1.5^2), and misfitting items that respond to a second dimension. Each dataset was checked against Part A's cached infit.
- **Conditions.** Null; one misfitting item at rho = .50; three misfitting items at rho = .15 (the strongest cascade) and at rho = .75 (subtle misfit).
- **Sample sizes.** 500, 2,000, 5,000. 200 datasets per cell.
- **FWB.** 200 draws per dataset, exponential weights normalised to mean 1, weighted CML refit with `psychotools::pcmodel()`, infit from the conditional moments with the same weights.
- **Intervals.** Per item, the 2.5 and 97.5 percentiles of the draws. Simultaneous: in each draw take the largest standardised departure over items, $\max_i |\ \mathrm{MSQ}_{bi} - \mathrm{MSQ}_i\ | / \mathrm{SD}_i$, and use its 95th percentile c as one constant for all items, giving $\mathrm{MSQ}_i \pm c \cdot \mathrm{SD}_i$.
- **Target.** Each item's population MSQ, its Part A mean at n = 20,000 (300 datasets). Its Monte Carlo SE is about .0005.

The Monte Carlo SE of a 95 percent coverage rate is 1.5 points for one item over 200 datasets. Rates pooled over items are more precise but not independent, since items share datasets.

Checks

```
d |>
  group_by(cond, n) |>
  summarise(datasets = n_distinct(rep),
            max_match_error = signif(max(match_err), 2),
            min_draws_ok = min(draws_ok), .groups = "drop") |>
  kb()
```

cond	n	datasets	max_match_error	min_draws_ok
null	500	200	0	200
null	2000	200	0	200
null	5000	200	0	200
xdim50_k1	500	200	0	200
xdim50_k1	2000	200	0	200
xdim50_k1	5000	200	0	200
xdim15_k3	500	200	0	200
xdim15_k3	2000	200	0	200
xdim15_k3	5000	200	0	200
xdim75_k3	500	200	0	200
xdim75_k3	2000	200	0	200
xdim75_k3	5000	200	0	200

0 datasets failed. About 45 minutes on 10 cores.

Coverage, item by item

Share of items whose interval contains the population MSQ (%), by kind of item. Nominal is 95.

```
d |>
  group_by(role, cond, n) |>
  summarise(percentile = 100 * mean(pop >= lo & pop <= hi),
            simultaneous = 100 * mean(pop >= slo & pop <= shi),
            .groups = "drop") |>
  pivot_wider(names_from = n, values_from = c(percentile, simultaneous),
              names_glue = "{.value} n={n}") |>
  kb(digits = 1)
```

role	cond	percentile n=500	percentile n=2000	percentile n=5000	simultaneous n=500	simultaneous n=2000	simultaneous n=5000
null	null	94.2	94.8	94.5	99.7	99.6	99.8
fitting, others misfit	xdim50_k1	93.8	93.6	94.0	99.7	99.5	99.6
fitting, others misfit	xdim15_k3	94.2	94.0	93.8	99.6	99.7	99.7

role	cond	per-centile n=500	per-centile n=2000	per-centile n=5000	simul-taneous n=500	simultane-ous n=2000	simultane-ous n=5000
fitting, oth-ers misfit	xdim75_k3	93.8	93.6	94.1	99.6	99.7	99.8
misfitting	xdim50_k1	94.5	95.0	90.5	100.0	99.5	99.0
misfitting	xdim15_k3	93.7	92.0	93.7	100.0	100.0	100.0
misfitting	xdim75_k3	94.3	95.0	92.8	99.7	99.7	99.8

Coverage of the misfitting items one by one, since they differ in targeting (0, −1 and −2 logits in the three-item conditions).

```
d |>
  filter(misfit) |>
  group_by(cond, item, n) |>
  summarise(pop = mean(pop),
            percentile = 100 * mean(pop >= lo & pop <= hi),
            .groups = "drop") |>
  pivot_wider(names_from = n, values_from = percentile,
              names_prefix = "n = ") |>
  kb(digits = c(0, 0, 3, 1, 1, 1))
```

cond	item	pop	n = 500	n = 2000	n = 5000
xdim50_k1	11	1.377	94.5	95.0	90.5
xdim15_k3	1	1.272	95.5	89.5	92.0
xdim15_k3	6	1.374	95.5	92.5	94.0
xdim15_k3	11	1.432	90.0	94.0	95.0
xdim75_k3	1	1.093	93.0	95.5	94.0
xdim75_k3	6	1.122	94.5	94.5	94.0
xdim75_k3	11	1.143	95.5	95.0	90.5

Simultaneous coverage

Share of datasets in which every item's simultaneous interval contains its population MSQ (%). Nominal is 95. The mean of c is shown for reference (1.96 would be the per-item constant).

```
d |>
  group_by(cond, n, rep) |>
  summarise(all = all(pop >= slo & pop <= shi), cmax = first(cmax),
            .groups = "drop") |>
  group_by(cond, n) |>
  summarise(all_covered = 100 * mean(all), mean_c = mean(cmax),
            .groups = "drop") |>
  kb(digits = c(0, 0, 1, 2))
```

cond	n	all_covered	mean_c
null	500	93.0	3.05
null	2000	92.5	3.00

cond	n	all_covered	mean_c
null	5000	95.5	2.98
xdim50_k1	500	93.5	3.07
xdim50_k1	2000	91.0	3.00
xdim50_k1	5000	92.0	2.98
xdim15_k3	500	94.0	3.06
xdim15_k3	2000	95.0	2.99
xdim15_k3	5000	94.5	2.98
xdim75_k3	500	91.5	3.06
xdim75_k3	2000	93.5	2.99
xdim75_k3	5000	95.5	2.99

Where misses come from

Coverage can fail because the observed MSQ is biased at that n, or because the FWB spread is too narrow. The table compares the mean observed MSQ with the population value, and the mean FWB SD with the SD of the observed MSQ over datasets, which is its real sampling SD at that n.

```
d |>
  group_by(role, cond, n, item) |>
  summarise(bias = mean(obs) - first(pop), true_sd = sd(obs),
            fwb_sd = mean(fwb_sd), .groups = "drop") |>
  group_by(role, cond, n) |>
  summarise(mean_abs_bias = mean(abs(bias)), max_abs_bias = max(abs(bias)),
            sd_ratio = mean(fwb_sd / true_sd), .groups = "drop") |>
  kb(digits = c(0, 0, 0, 4, 4, 2))
```

role	cond	n	mean_abs_bias	max_abs_bias	sd_ratio
null	null	500	0.0029	0.0060	1.00
null	null	2000	0.0017	0.0042	1.02
null	null	5000	0.0010	0.0021	1.01
fitting, others misfit	xdim50_k1	500	0.0026	0.0069	1.00
fitting, others misfit	xdim50_k1	2000	0.0013	0.0039	1.00
fitting, others misfit	xdim50_k1	5000	0.0010	0.0029	1.00
fitting, others misfit	xdim15_k3	500	0.0023	0.0062	1.01
fitting, others misfit	xdim15_k3	2000	0.0014	0.0031	1.01
fitting, others misfit	xdim15_k3	5000	0.0011	0.0028	0.99
fitting, others misfit	xdim75_k3	500	0.0028	0.0070	0.99
fitting, others misfit	xdim75_k3	2000	0.0018	0.0042	0.99
fitting, others misfit	xdim75_k3	5000	0.0008	0.0022	1.02
misfitting	xdim50_k1	500	0.0031	0.0031	1.05
misfitting	xdim50_k1	2000	0.0020	0.0020	1.08
misfitting	xdim50_k1	5000	0.0008	0.0008	0.90

role	cond	n	mean_abs_bias	max_abs_bias	sd_ratio
misfitting	xdim15_k3	500	0.0021	0.0030	1.00
misfitting	xdim15_k3	2000	0.0018	0.0023	0.97
misfitting	xdim15_k3	5000	0.0007	0.0016	0.98
misfitting	xdim75_k3	500	0.0038	0.0041	1.02
misfitting	xdim75_k3	2000	0.0010	0.0023	1.03
misfitting	xdim75_k3	5000	0.0005	0.0013	0.98

A ratio near 1 means the FWB spread matches the real one.

The floor in practice

Decisions with the simultaneous intervals against a band $[1-\delta, 1+\delta]$, for two illustrative values of δ. These are not recommendations, since δ should come from impact. An item's population value is "within tolerance" when $|\ \mathrm{MSQ} - 1\ | < \delta$. The table gives the share of items with each decision (%).

```
expand_grid(delta = c(0.05, 0.10)) |>
  group_by(delta) |>
  group_modify(function(x, key) {
    dl <- key$delta
    d |>
      mutate(truth = ifelse(abs(pop - 1) < dl, "within tolerance",
                            "beyond tolerance"),
             decision = case_when(slo >= 1 - dl & shi <= 1 + dl ~ "inside",
                                  shi < 1 - dl | slo > 1 + dl ~ "outside",
                                  TRUE ~ "undecided"))
  }) |>
  ungroup() |>
  count(delta, cond, role, truth, n, decision, name = "k") |>
  group_by(delta, cond, role, truth, n) |>
  mutate(items = sum(k), pct = 100 * k / sum(k)) |>
  select(-k) |>
  pivot_wider(names_from = decision, values_from = pct, values_fill = 0) |>
  mutate(delta = sprintf("%.2f", delta)) |>
  select(delta, cond, role, truth, n, items, any_of(c("inside", "undecided",
                                                      "outside"))) |>
  arrange(delta, cond, role, truth, n) |>
  kb(digits = 1)
```

delta	cond	role	truth	n	items	in-side	unde-cided	out-side
0.05	null	null	within tolerance	500	4000	0.0	100.0	0.0
0.05	null	null	within tolerance	2000	4000	0.0	100.0	0.0
0.05	null	null	within tolerance	5000	4000	11.6	88.4	0.0
0.05	xdim50_k1	fitting, others misfit	within tolerance	500	3800	0.0	99.9	0.1

delta	cond	role	truth	n	items	inside	undecided	outside
0.05	xdim50_k1	fitting, others misfit	within tolerance	2000	3800	0.0	100.0	0.0
0.05	xdim50_k1	fitting, others misfit	within tolerance	5000	3800	6.2	93.8	0.0
0.05	xdim50_k1	misfitting	beyond tolerance	500	200	0.0	2.0	98.0
0.05	xdim50_k1	misfitting	beyond tolerance	2000	200	0.0	0.0	100.0
0.05	xdim50_k1	misfitting	beyond tolerance	5000	200	0.0	0.0	100.0
0.05	xdim15_k3	fitting, others misfit	beyond tolerance	500	3400	0.0	98.8	1.2
0.05	xdim15_k3	fitting, others misfit	beyond tolerance	2000	3400	0.0	96.3	3.7
0.05	xdim15_k3	fitting, others misfit	beyond tolerance	5000	3400	0.0	90.2	9.8
0.05	xdim15_k3	misfitting	beyond tolerance	500	600	0.0	4.7	95.3
0.05	xdim15_k3	misfitting	beyond tolerance	2000	600	0.0	0.0	100.0
0.05	xdim15_k3	misfitting	beyond tolerance	5000	600	0.0	0.0	100.0
0.05	xdim75_k3	fitting, others misfit	within tolerance	500	3400	0.0	99.9	0.1
0.05	xdim75_k3	fitting, others misfit	within tolerance	2000	3400	0.0	100.0	0.0
0.05	xdim75_k3	fitting, others misfit	within tolerance	5000	3400	7.0	93.0	0.0
0.05	xdim75_k3	misfitting	beyond tolerance	500	600	0.0	97.2	2.8
0.05	xdim75_k3	misfitting	beyond tolerance	2000	600	0.0	63.2	36.8
0.05	xdim75_k3	misfitting	beyond tolerance	5000	600	0.0	26.5	73.5
0.10	null	null	within tolerance	500	4000	0.0	100.0	0.0
0.10	null	null	within tolerance	2000	4000	64.0	36.0	0.0
0.10	null	null	within tolerance	5000	4000	99.8	0.2	0.0
0.10	xdim50_k1	fitting, others misfit	within tolerance	500	3800	0.0	100.0	0.0

delta	cond	role	truth	n	items	in-side	unde-cided	out-side
0.10	xdim50_k1	fitting, others misfit	within toler-ance	2000	3800	56.1	43.9	0.0
0.10	xdim50_k1	fitting, others misfit	within toler-ance	5000	3800	98.2	1.8	0.0
0.10	xdim50_k1	misfitting	beyond toler-ance	500	200	0.0	8.5	91.5
0.10	xdim50_k1	misfitting	beyond toler-ance	2000	200	0.0	0.0	100.0
0.10	xdim50_k1	misfitting	beyond toler-ance	5000	200	0.0	0.0	100.0
0.10	xdim15_k3	fitting, others misfit	within toler-ance	500	3400	0.0	100.0	0.0
0.10	xdim15_k3	fitting, others misfit	within toler-ance	2000	3400	6.9	93.1	0.0
0.10	xdim15_k3	fitting, others misfit	within toler-ance	5000	3400	31.0	69.0	0.0
0.10	xdim15_k3	misfitting	beyond toler-ance	500	600	0.0	17.5	82.5
0.10	xdim15_k3	misfitting	beyond toler-ance	2000	600	0.0	0.0	100.0
0.10	xdim15_k3	misfitting	beyond toler-ance	5000	600	0.0	0.0	100.0
0.10	xdim75_k3	fitting, others misfit	within toler-ance	500	3400	0.0	100.0	0.0
0.10	xdim75_k3	fitting, others misfit	within toler-ance	2000	3400	57.2	42.8	0.0
0.10	xdim75_k3	fitting, others misfit	within toler-ance	5000	3400	98.6	1.4	0.0
0.10	xdim75_k3	misfitting	beyond toler-ance	500	400	0.0	99.8	0.2
0.10	xdim75_k3	misfitting	beyond toler-ance	2000	400	0.0	97.2	2.8
0.10	xdim75_k3	misfitting	beyond toler-ance	5000	400	0.0	82.0	18.0
0.10	xdim75_k3	misfitting	within toler-ance	500	200	0.0	100.0	0.0
0.10	xdim75_k3	misfitting	within toler-ance	2000	200	1.0	99.0	0.0
0.10	xdim75_k3	misfitting	within toler-ance	5000	200	0.5	99.5	0.0

For comparison, the share of fitting items that a test against MSQ = 1 would flag with these same simultaneous intervals (interval excludes 1), in the conditions where other items misfit:

```
d |>
  filter(role == "fitting, others misfit") |>
  group_by(cond, n) |>
  summarise(flagged_vs_1 = 100 * mean(slo > 1 | shi < 1), .groups = "drop") |>
  pivot_wider(names_from = n, values_from = flagged_vs_1,
              names_prefix = "n = ") |>
  kb(digits = 1)
```

cond	n = 500	n = 2000	n = 5000
xdim50_k1	1.1	3.2	7.4
xdim15_k3	11.4	58.9	94.0
xdim75_k3	1.1	3.2	7.6

Interpretation

Written 2026-10-12 after the run. 2,400 datasets, none failed, every dataset identical to Part A (largest difference in infit 1.3e-13).

1. The FWB spread matches the real sampling spread. The mean FWB SD over the SD of the observed MSQ across datasets is 0.97 to 1.03 in every cell with many items behind it, for null, cascade-affected and misfitting items alike. The two single-item cells (one misfitting item at rho = .50) give 0.90 to 1.08, where the empirical SD itself rests on 200 values. The observed MSQ is nearly unbiased for its population value, with a mean absolute bias of .001 to .004 and at most .007. So FWB does what the floor needs: it measures how uncertain the observed MSQ is, under the data as they are.

2. Per-item coverage is close to nominal, slightly under. The percentile interval covers the population MSQ for 93.6 to 95.0 percent of items in the pooled rows, and 89.5 to 95.5 percent for single misfitting items, where the Monte Carlo SE is 1.5 points. The shortfall of about one point is consistent with estimating 2.5 and 97.5 percentiles from 200 draws, which makes the interval slightly too short. More draws, or an SD-based interval (observed MSQ plus or minus 1.96 FWB SDs), would remove most of it.

3. Simultaneous coverage is close to nominal. The share of datasets with every item covered is 91.0 to 95.5 percent (mean about 93.5). The constant c is about 3.0 at every n, close to the Bonferroni value for 20 items (3.02), because the items' departures are nearly independent across draws. With c from 200 draws its 95th percentile is noisy, and B = 1,000 would be the first thing to try before calling the shortfall real.

4. The floor stops the cascade. With three strongly misfitting items (rho = .15), the fitting items have population MSQ about 0.92. A test against MSQ = 1, using these same simultaneous intervals, flags them at 11, 59 and 94 percent at n = 500, 2,000 and 5,000. With $\delta = .10$ the floor flags none of them at any n. They are mostly undecided, because 0.92 lies near the boundary at 0.90, and the share cleared as inside grows with n (0 to 31 percent). The misfitting items in the same datasets are declared outside in 83, 100 and 100 percent.

5. The floor turns large n into a positive finding. Under a true Rasch model, with $\delta = .10$, items are cleared as practically fitting in 0, 64 and 99.8 percent of cases at n = 500, 2,000 and 5,000. No null item was ever declared outside. A test cannot give this kind of result.

6. Small misfit near the boundary stays undecided. At rho = .75 the misfitting items have population MSQ 1.09 to 1.14. With $\delta = .10$, the two above 1.10 are declared outside in 0.2, 2.8 and 18 percent and are otherwise undecided. The one at 1.09 is never cleared and is falsely declared outside in at most 1 percent. Undecided is the honest answer for an item within a few hundredths of the tolerance.

7. The choice of δ sets the sample size the floor needs. With $\delta = .05$ the simultaneous half-width (about 3 SDs, .048 at n = 5,000) is as wide as the band, so even null items are cleared only 12 percent

of the time at n = 5,000, and the cascade items at 0.92 count as beyond tolerance and are declared outside in up to 10 percent. A narrower band needs a larger sample before it can clear anything. For 20 dichotomous items, n = 5,000 supports a band of about $\pm.10$ and n = 2,000 only partly.

Limits. Dichotomous items, 20 items, normal theta, second-dimension misfit only. B = 200 draws. The values of δ are illustrations, not recommendations. In use, δ should come from impact, and the shortfall in simultaneous coverage should be rechecked with more draws.